Optimized Composition and Manufacturing Method of Natural Rubber-EPDM Composites with Nano Oxides: Analytical Data and Selection Criteria

5. Analytical Data Supporting Nano-Oxide Selection

5.1 Nano-Zinc Oxide: Cure Activation Efficiency

The selection of nano-ZnO as the primary curing activator in NR/EPDM systems is supported by quantitative cure kinetics data. Spherical and elongated ZnO nanoparticles synthesized by homogeneous precipitation were characterized by TEM and XRD, confirming impurity-free particles with dimensions ranging from 10 to 40 nm . When incorporated into NR/recycled EPDM blends, these nanoparticles produced a **better state of curing and higher maximum torque** with a **significant decrease in optimum curing time and scorching time** compared to conventional ZnO .

The mechanistic basis for this enhanced efficiency lies in the increased surface area of nanoparticles, which provides more effective activation of the vulcanization process. The study confirmed that **a low content of ZnO nanoparticles was sufficient to give higher mechanical properties compared to formulations containing a higher content of conventional ZnO** . This dose-response relationship demonstrates that nano-ZnO achieves equivalent or superior cure activation at reduced loading levels, which is economically and environmentally advantageous.

Thermo-mechanical analysis further revealed that ZnO nanoparticles not only act as a curing activator but also improve the **thermal stability and dynamic mechanical properties** of NR/EPDM blends . The cure kinetic analysis of EPDM rubber containing ZnO nanoparticles showed that the **activation energy of the curing reaction was remarkably reduced** upon decreasing ZnO particle size . This quantitative evidence establishes nano-ZnO as the most efficient curing activator among the nano-oxides evaluated.

5.2 Nano-Silica: Mechanical Reinforcement and Swelling Resistance

The selection of nano-silica as the primary mechanical reinforcement is supported by systematic studies on NR/EPDM (70/30) blends using melt compounding with nanosilica derived from rice husk ash . The influence of modified nanosilica on curing, mechanical, and swelling resistance properties was examined using two modifiers: TESPT (bis(3-triethoxysilylpropyl)tetrasulfide) and TWEEN-20 .

The analytical data demonstrate that **modified nanosilica improves mechanical properties and swelling resistance** of NR/EPDM blends . The reinforcement mechanism involves enhanced filler-rubber interaction through silane coupling agents, which establishes covalent bonding between the silica surface and the rubber matrix. This covalent silica-polymer network is **more stable than the carbon black-polymer network**, resulting in a **reduced rate of bond breaking and reformation during deformation cycles** and consequently lower loss modulus .

Comparative analysis of silica versus carbon black reveals that silica exhibits **better filler-filler interactions but poorer filler-polymer interactions** when used without coupling agents . However, with proper silane modification, silica imparts **distinct dynamic mechanical properties and a stronger reinforcing effect** . The following characteristics are primarily improved when silica and a coupling agent replace carbon black: tear resistance, abrasion resistance, heat resistance, hardness, flex stability, modulus, stiffness, tack, resilience, and heat buildup . The superior reinforcing effect of modified silica enables **reduced filler content without adversely affecting the property profile** .

5.3 Nano-Alumina: Thermal Conductivity Enhancement

The selection of nano-alumina for thermal management applications is supported by quantitative thermal conductivity data. EPDM/Al_2O_3 composites were prepared with varying Al_2O_3 content, and thermal conductivity was measured systematically . As Al_2O_3 loading increased, **mechanical properties became more excellent and thermal conductivity increased significantly** .

At 100 phr loading, the thermal conductivity of the composite reached **0.53 W/(m·K)**, which is approximately **32.3% higher than carbon black N550-filled composites** and **57.7% higher than PVC-filled composites** . The compression fatigue value increased only

slightly, indicating that nano-Al_2O_3 filler **greatly reduces friction between filler and rubber matrix**, resulting in **smaller heat accumulation** .

SEM analysis confirmed that Al_2O_3 filler **disperses uniformly in the rubber matrix**, and the composite exhibits a **well-developed crosslinked network**, which is significant for extending service life . When in-situ modified with the silane coupling agent Si69, the mechanical properties (especially dynamic mechanical properties) of nano-Al_2O_3 filled composites were **obviously improved without influencing thermal conductivity** . Compared to traditional reinforcing fillers (carbon black N330 and silica), in-situ modified nano-Al_2O_3 filled composites exhibit **excellent performance in mechanical properties as well as better thermal conductivity, especially lower compression heat build-up and better fatigue resistance** . This indicates that nano-Al_2O_3 is suitable for preparing rubber products serving in dynamic conditions with longer expected service life .

5.4 Nano-Titania: Thermal Stability and Phase-Specific Effects

The analytical data for nano-titania reveal complex effects on NR/EPDM blends that require careful interpretation for material selection. TEM and XRD results show that **titania particles are concentrated in the NR phase**, and the presence of titania in EPDM **seems to initiate the formation of crystals in the rubber phase** . This phase-selective localization is a critical consideration for blend design.

However, DMA and tensile testing results reveal that **titania particles inhibited the vulcanization process in both rubbers** . The mechanical properties of the nanocomposites were influenced by three competing factors: the interaction between nanoparticles and rubber, the reduced vulcanization of both rubber phases, and the development of crystallinity in EPDM .

TGA results demonstrate **improved thermal stability of EPDM, and of the EPDM phase in the blend, for samples containing titania nanoparticles** . This thermal stability enhancement is attributed to the barrier effect of titania particles, which impede the diffusion of degradation products and volatile species through the rubber matrix.

The inhibitory effect on vulcanization necessitates adjustment of the cure system when nano-titania is incorporated. The selection of nano-titania should therefore be limited to applications where **thermal stability enhancement outweighs the need for maximum crosslink density**, and where the cure system can be optimized to compensate for the reduced vulcanization rate.

5.5 Comparative Selection Matrix

Based on the analytical data, the following selection criteria emerge for nano-oxide additives in NR/EPDM composites:

Nano-Oxide	Primary Function	Key Analytical Evidence	Recommended Loading	Limitation
Nano-ZnO	Cure activation	Reduced optimum cure time, lower activation energy, higher mechanical properties at low loading	2–4 phr	None significant
Nano-Silica	Mechanical reinforcement	Improved tensile strength, swelling resistance, dynamic properties with TESPT modification	10 phr	Requires silane coupling agent
Nano-Al_2O_3	Thermal conductivity	0.53 W/(m·K) at 100 phr, 32.3% higher than N550 carbon black	5–100 phr	High loading needed for thermal effect
Nano-TiO_2	Thermal stability	Improved EPDM thermal stability, but inhibits vulcanization	2–5 phr	Cure system adjustment required

5.6 Synergistic Considerations and Final Selection Rationale

The analytical data support a **dual-function nano-oxide strategy** rather than relying on a single oxide. Nano-ZnO should be selected as the **cure activator** because it provides superior cure kinetics and mechanical properties at reduced loading . Nano-silica should be selected as the **primary mechanical reinforcement** because it delivers the most significant improvements in tensile strength and swelling resistance when properly modified . For applications requiring thermal management, nano-Al_2O_3 can be added as a **secondary filler** because it enhances

thermal conductivity without compromising mechanical performance when in-situ modified with Si69 .

The selection of nano-TiO_2 should be restricted to applications where **thermal stability of the EPDM phase is the paramount requirement**, and where the cure system can be adjusted to compensate for its inhibitory effect on vulcanization .

The final optimized composition for general-purpose NR/EPDM nanocomposites therefore comprises: 70 phr NR, 30 phr EPDM, 4 phr nano-ZnO (cure activator), 10 phr modified nano-silica (mechanical reinforcement), 30 phr carbon black N330 (co-reinforcement), and conventional cure system with TBBS/MBTS accelerators. For thermally demanding applications, 5–10 phr nano-Al_2O_3 is additionally incorporated as a thermally conductive filler. This composition is supported by the quantitative analytical data demonstrating that nano-ZnO reduces cure time and activation energy , nano-silica improves mechanical and swelling properties , and nano-Al_2O_3 enhances thermal conductivity with minimal impact on dynamic fatigue .

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